Smart Home Safety System

Automated IoT Monitoring & Response Platform

Distributed and Internet of Things Software Architectures
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Problem Context

Safety Challenges in Modern Smart Homes

Intrusion Detection
Unauthorized entry when residents away

Fire Hazard
High temperature and smoke detection

Gas Leak
Abnormal consumption spike detection

Solution

- ✓ Real-time monitoring
- ✓ Automated threat detection
- ✓ Immediate safety responses
- ✓ Distributed IoT architecture
- ✓ Event logging & audit trail

System Architecture

Observers & Apps

HTTP/REST

↓

Data Collector & Manager

FastAPI + MQTT Client

Rule Engine | Device Registry | State Store

MQTT Broker

↓

Sensors

Door/Window
Environment

Actuators

Alarm | Switch | Sprinkler

Hybrid Devices

Light | Gas | Electricity | Water

Architecture Pattern

Distributed Event-Driven Architecture

Scalability
Add new devices without modifying central manager. Each device operates independently.

Real-time Response
Events processed immediately as they occur. Critical for safety applications.

Loose Coupling
Devices communicate through message broker, unaware of each other. Failure isolation.

Industry Standard
Aligns with AWS IoT, Azure IoT Hub, Google Cloud IoT best practices.

Publish/Subscribe Pattern: Devices publish events → Manager reacts → Commands dispatched

Communication Protocols

MQTT (Device ↔ Manager)

- Lightweight & Efficient
- Publish/Subscribe pattern
- QoS levels for reliability
- Persistent connections
- Industry standard (Eclipse Paho)

Topic: home/<home_id>/<device_id>/<channel>

HTTP/REST (Observer ↔ Manager)

- Ubiquitous support
- Stateless operations
- Cacheable responses
- Request/response pattern
- Easy integration (FastAPI)

Endpoints: /status, /config, /devices

Data Format: JSON

Human-readable | Language-agnostic | Flexible schema | Native FastAPI support

IoT Device Ecosystem

10 Emulated Devices Across 3 Categories

Category	Device	Data/Command
Sensors (3)	door_1, window_1 env_1	open: bool temperature, pm10: float
Actuators (3)	alarm_controller alarm_switch sprinkler	cmd: set(on: bool) cmd: set(armed: bool) cmd: set(on: bool)
Hybrid (4)	mobile_light gas_meter electricity_meter water_meter	telemetry: energy_kwh, on, level + cmd: set(on: bool, level: str) telemetry: total, delta, supply_on + cmd: set(supply_on: bool) " "

All devices publish to MQTT topic: home/home_1/<device_id>/{telemetry|state|cmd}

Safety Rule Engine

Rule 1: Intrusion
Condition:
armed=True + door open
Actions:
Alarm ON + Light HIGH

Rule 2: Fire
Condition:
temp≥60°C + pm10≥150
Actions:
Alarm ON + Sprinkler ON

Rule 3: Gas Leak
Condition:
delta/prev ≥ 2.0x spike
Actions:
Alarm ON + Gas shutoff

Key Features

Edge Detection
Rules trigger only on False→True transition, preventing duplicate activations

Cooldown Period
5-second buffer prevents command spam during persistent conditions

Configurable Thresholds
All parameters adjustable via REST API (temp, pm10, gas_spike_ratio)

Event Logging
Complete audit trail in JSONL format: outputs/events.log

Telemetry → Evaluate Rules → Trigger Actions → Log Events → Dispatch Commands

Implementation Highlights

Technology Stack

Backend: Python 3.10+ | FastAPI | Uvicorn
MQTT: Eclipse Paho | Mosquitto Broker
Data: Pydantic models | Thread-safe stores
Code: 950+ lines across 9 Python modules

Key Design Decisions

- Thread-safe state management
- Alarm switch via telemetry (bidirectional control)
- Hybrid device pattern (sensor + actuator)
- Single-home design (scalable to multi-tenant)

REST API Endpoints

Endpoint	Method	Purpose
/status	GET	Complete system snapshot (config, devices, telemetry, state)
/config	GET / PUT	Retrieve or update system configuration (thresholds, rules)
/devices	GET / POST / DELETE	Device registry CRUD operations

✓ All 3 rules validated | ✓ Edge detection verified | ✓ Demo script included

Demo & Results

Automated Demo Script

1. Arm system → Trigger intrusion → Alarm + Light
2. High temp/PM10 → Trigger fire → Alarm + Sprinkler
3. Gas spike → Trigger gas leak → Alarm + Shutoff

All events logged to outputs/events.log

Key Achievements

- ✓ Functional IoT safety system
- ✓ 10 devices, 3 automated rules
- ✓ Distributed event-driven architecture
- ✓ MQTT + REST communication
- ✓ Thread-safe, production-ready code

Future Enhancements

- Multi-home support
- Persistent storage (database)
- Web dashboard UI
- Cloud integration (AWS IoT)
- Machine learning anomaly detection

Thank You!